

$$x^{(i)} \longrightarrow y^{(i)}$$

$$y(x) = \varphi_{\beta}(x)$$

$$2500 = y(20) = \varphi_{\beta}(20)$$

x	y
20	2,500
25	2,500
20	1,800
30	3,600

$\uparrow$ horas $\uparrow$ salario

$$\varphi_{\beta}(x) = \varphi(\overbrace{x}^{\text{variable}}, \overbrace{\beta}^{\text{constante}})$$

modelo generalizado

$$\beta \rightarrow \varphi$$

$$y^{(i)} \stackrel{T}{=} x^{(i)}$$

(nube de datos)

$$2500 \stackrel{T}{=} 20$$

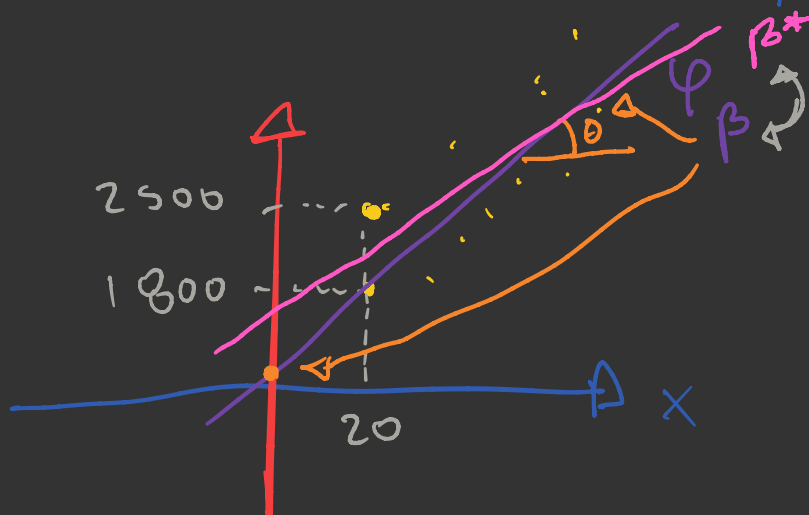
→ parejas observadas

$$2500 \stackrel{T}{=} 25$$

$$(20, 2500)$$

$$(25, 2500)$$

$$(20, 1800)$$



$$m = \tan \theta$$

$$\beta_1$$

$$b$$

$$\beta_0$$

$$\hat{y} = \varphi\left(\underbrace{x}_c, \underbrace{\beta}_v\right)$$

Principio del Perceptrón Simple

- Modelo generalizado
(Modelo del predictor)

$$z = \underbrace{x \cdot \beta}_{\text{predictor}}$$

$$\varphi(x, \beta) = x \cdot \beta$$

- Error y Pérdida

$$\varepsilon = y - \hat{y}$$

$$(-\infty, \infty) \quad \varphi(x, \beta) = x \cdot \beta \quad \left\{ \begin{array}{l} \text{lineal} \\ \text{kernel} \\ = \sigma(x \cdot \beta) \end{array} \right.$$

~~$$L(\varepsilon) = \varepsilon$$~~

$$L(\varepsilon) = \varepsilon^2$$

$$L(\varepsilon) = \frac{1}{2} \varepsilon^2 \quad \text{"} \rightarrow \infty$$

MSE



siempre es positivo
" suave

$$\frac{1}{2} |\varepsilon| \quad \rightarrow \infty$$

MAE



! no diferenciable

↓
problema en la optimización

$$L(\varepsilon) \rightarrow 0$$

$$\beta^* - \beta = 0 \quad \beta^* = \beta \quad \text{"}$$

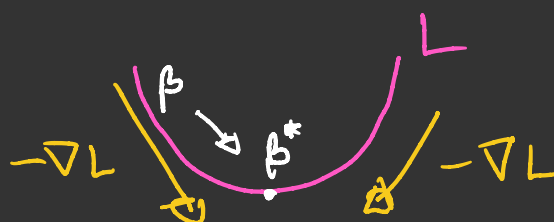
- Optimizador

• - Ingenua (Bayesiana / Estocástica) ~~✗~~

✓ - Gradiente Descenso

- Newton / Raphson / BFGS / ADAM / ... $\nabla \rightarrow H(\nabla^2)$

$$\rightarrow \Delta \beta = \beta^* - \beta \propto -\nabla_{\beta} L(\varepsilon) \quad \therefore$$

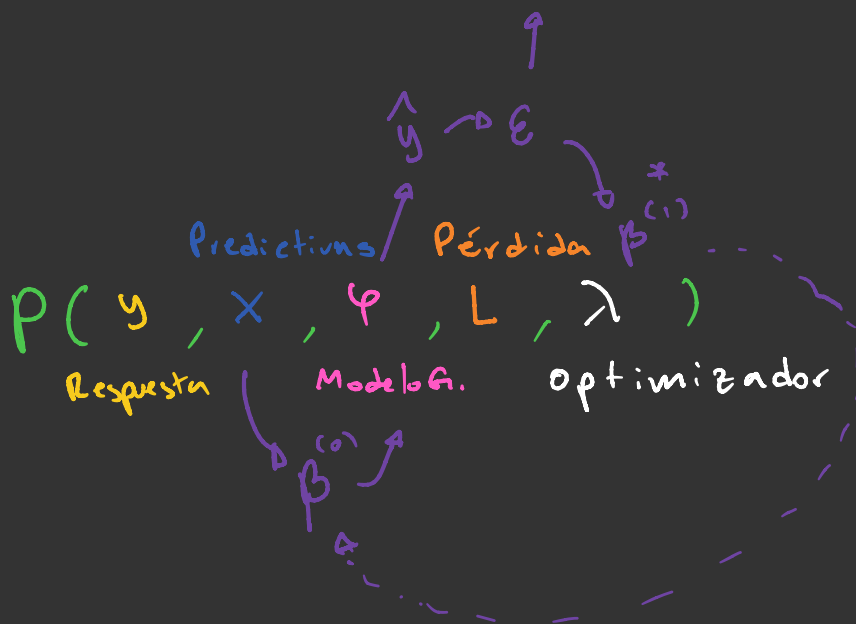


$$\text{Perceptrón}(x, y; \varphi, L, \lambda) \rightarrow \hat{y}^*$$

↳ Neurona Artificial

$$|\epsilon| \geq 10^{-3}$$

$\epsilon \rightarrow 0$ " / MÁXITER.



Modelo Linear Generalizado

$$y^{(i)} = \varphi(x^{(i)}, \beta^{(0)}) \quad \beta^{(0)} = (\beta_0^{(0)}, \beta_1^{(0)})$$

$$y^{(i)} = \beta_0^{(0)} + \beta_1^{(0)} x^{(i)} \quad \beta^{(0)} = (-0.3, 0.5)$$

$$y^{(i)} = -0.3 + 0.5 x^{(i)}$$

$$2500 = -0.3 + 0.5 (20)$$

$$2500 = -0.3 + 10$$

$$2500 = 9.7$$

$$\hat{y}(x^{(i)}) = \hat{y}(20) = 9.7$$

x	y
...	...
i. 20	2,500
...	...

$$\epsilon = 2500 - 9.7 = 2490.3$$

$$L(\epsilon) = \frac{1}{2} \epsilon^2 = 3,100,797.045$$

$$\beta^{(1)} = \beta^{(0)} + \eta \epsilon \nabla_{\beta} \varphi$$

Gradiente Descenso

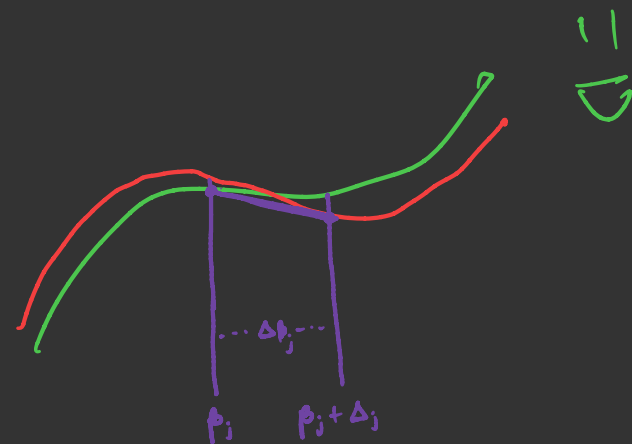
$$\nabla_z f(z) = \left(\frac{\partial f}{\partial z_0}, \frac{\partial f}{\partial z_1} \right)$$

$$\nabla_{\beta} \varphi(\beta)$$

$$X - \text{Edad} \mid \beta_0, \beta_1$$

$$X - \text{Edad}, \text{Altura} \mid \beta_0, \beta_1, \beta_2$$

$$\nabla_{\beta} \varphi(\beta) = \left(\frac{\partial \varphi}{\partial \beta_0}, \frac{\partial \varphi}{\partial \beta_1}, \frac{\partial \varphi}{\partial \beta_2} \right)$$



$$\frac{\partial \varphi}{\partial \beta_j} = \lim_{\Delta \beta_j \rightarrow 0} \frac{\varphi(x, \beta_0, \dots, \beta_j + \Delta \beta_j, \dots) - \varphi(x, \beta_0, \dots, \beta_j, \dots)}{\Delta \beta_j}$$

$$\beta^{(0)} \rightarrow \beta^{(1)}$$

$$y = \beta_0^{(t)} + \beta_1^{(t)} x_1 + \dots + \beta_k^{(t)} x_k$$

$$\Delta \beta = \beta^{(t+1)} - \beta^{(t)}$$

$$\beta^{(t)} \rightarrow \varepsilon$$

$$\varepsilon = y - \hat{y}$$

$$\hat{y} = \varphi(x, \beta^{(t)})$$

$$\hat{y} = x \cdot \beta^{(t)}$$

$$L(\varepsilon) = \frac{1}{2} \varepsilon^2$$

$$L(\varepsilon) \rightarrow 0, \varepsilon \rightarrow 0$$

$$0 = y - \hat{y}$$

$$y = \hat{y}$$

$$\min_{\beta} L(\varepsilon)$$

$$\varepsilon(\beta) = y - \hat{y}$$

$$= y - \varphi(x, \beta)$$

$$= y - x \cdot \beta$$

$$\Delta \beta \propto -\nabla_{\beta} L$$

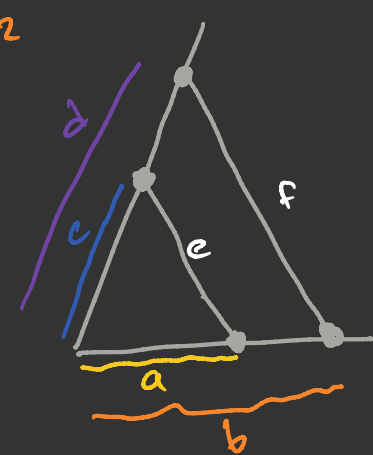
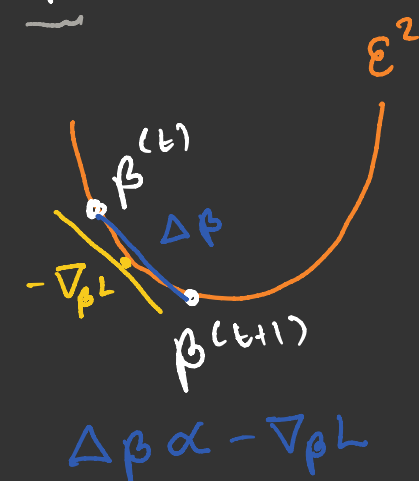


$$\Delta \beta = \eta (-\nabla_{\beta} L)$$

$$\beta^{(t+1)} - \beta^{(t)} = -\eta \nabla_{\beta} L$$

$$\beta^{(t+1)} = \beta^{(t)} - \eta \nabla_{\beta} L$$

$$\nabla_{\beta} L = -\nabla_{\beta} \varphi$$



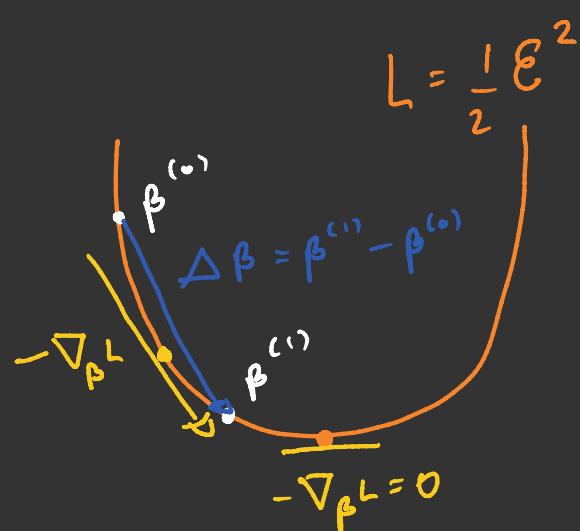
Triángulos semejantes

$$\frac{a}{c} \propto \frac{b}{d}$$

$$\frac{e}{a} \propto \frac{f}{b}$$

Receso

9:15 - 9:30 pm



$$\Delta \beta \propto -\nabla_{\beta} L$$

$$\Delta \beta = \eta (-\nabla_{\beta} L)$$

$$\beta^{(t+1)} - \beta^{(t)} = \eta (-\nabla_{\beta} L)$$

$$\beta^{(t+1)} = \beta^{(t)} + \eta (-\nabla_{\beta} L)$$

$$\beta^{(t+1)} = \beta^{(t)} - \eta \nabla_{\beta} L$$

$$-\nabla_{\beta} L = 0$$

$$\beta^* = \beta^{(T)} \quad \square$$

linear $\varphi = x \cdot \beta$

$$\nabla_{\beta} \varphi = x$$

$$\beta^{(t+1)} = \beta^{(t)} + \eta \varepsilon x$$

$$\beta^{(t+1)} = \beta^{(t)} + \eta (y - \hat{y}) x \quad \square$$

$$|\varepsilon| \approx 10^{-6} \quad \text{à} \quad t \approx T$$

$$\rightarrow \beta^*$$

$$\hat{y} = x \cdot \beta^*$$

no linear $\varphi = \sigma(x \cdot \beta)$

σ - sigmoïde (logística)

$$\nabla_{\beta} \varphi = x \sigma(x \cdot \beta) (1 - \sigma(x \cdot \beta))$$

$$\beta^{(t+1)} = \beta^{(t)} + \eta (y - \hat{y}) x \sigma(x \cdot \beta) (1 - \sigma(x \cdot \beta))$$

$$L(\varepsilon) = \frac{1}{2} \varepsilon^2$$

$$\nabla_{\beta} L = \frac{1}{2} (2\varepsilon) (\nabla_{\beta} \varepsilon)$$

$$\nabla_{\beta} L = \varepsilon \boxed{\nabla_{\beta} \varepsilon}$$

$$\nabla_{\beta} \varepsilon = \nabla_{\beta} (y - \hat{y})$$

$$\nabla_{\beta} \varepsilon = \nabla_{\beta} (y - \varphi(x, \beta))$$

$$\nabla_{\beta} \varepsilon = \nabla_{\beta} y - \nabla_{\beta} \varphi$$

$\underbrace{\nabla_{\beta} y}_{\text{constante}}$

$$\nabla_{\beta} \varepsilon = -\nabla_{\beta} \varphi$$

$$\nabla_{\beta} L = -\varepsilon \nabla_{\beta} \varphi$$

$$\beta^{(t+1)} = \beta^{(t)} - \eta \boxed{\nabla_{\beta} L}$$

$$\beta^{(t+1)} = \beta^{(t)} - \eta (-\varepsilon \nabla_{\beta} \varphi)$$

$$\beta^{(t+1)} = \beta^{(t)} + \eta \varepsilon \boxed{\nabla_{\beta} \varphi}$$

constante de aprendizaje (0.0001)

$$\sigma(z) = \frac{e^z}{1+e^z} = \frac{1}{1+e^{-z}}$$

$$\nabla_{\beta} \sigma(z) = \frac{\partial z}{\partial \beta} \frac{\partial \sigma(z)}{\partial z} = x \sigma(z) (1 - \sigma(z))$$

$\underbrace{\frac{\partial z}{\partial \beta}}_x \quad \underbrace{\frac{\partial \sigma(z)}{\partial z}}_{\sigma(z)(1-\sigma(z))}$

$z = x \cdot \beta \quad \frac{\partial (x \cdot \beta)}{\partial \beta} = x$

(x, y)

$$\varphi(x, \beta^{(0)}) = x \cdot \beta^{(0)} \quad \beta^{(0)} \approx \bar{0}$$

$$\hat{y} = x \cdot \beta^{(0)}$$

$$\varepsilon_0 = y - \hat{y}$$

$$L = \frac{1}{2} \varepsilon^2$$

$$\beta^{(1)} = \beta^{(0)} + \eta \varepsilon x$$

$$\varphi(x, \beta^{(1)}) = x \cdot \beta^{(1)}$$

$$\hat{y} = x \cdot \beta^{(1)}$$

$$\varepsilon_1 = y - \hat{y}$$

$$L = \frac{1}{2} \varepsilon^2$$

$$\beta^{(2)} = \beta^{(1)} + \eta \varepsilon x$$

$$\varphi(x, \beta^{(2)}) = x \cdot \beta^{(2)}$$

$$\hat{y} = x \cdot \beta^{(2)}$$

$$\varepsilon_2 = y - \hat{y}$$

$$L = \frac{1}{2} \varepsilon^2$$

